

Harmonizing Randomization Mechanisms for Differentially Private Federated Learning

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1 Mode Connectivity

Machine learning involves searching for a model parameter that minimizes a trainer-specified loss function [3]. We can visualize this process geometrically as moving on a loss surface to search the global minima. However, recovering the global minima becomes harder as the dimension of the model parameters increases, making it more likely that training terminates at a local minima for deep models [1, 5]. Recent research has shown that local minima on the loss surface (aka. *modes* of the loss function) are connected by simple curves, a phenomenon known as mode connectivity [6].

To find a curve between two modes, we perform a *curve finding procedure* similar to [6]. Let w_1 and w_2 be the model parameters of two trained networks in $\mathbb{R}^d$, and $\ell(w)$ be the cross-entropy loss function. Furthermore, let $\phi_\theta : [0, 1] \rightarrow \mathbb{R}^d$ be a continuous piece-wise smooth parametric curve, with parameters θ , such that the curve endpoints are $\phi_\theta(0) = w_1$ and $\phi_\theta(1) = w_2$. Our goal is to find the parameters θ that minimize the expectation of the loss $\ell(\phi_\theta(p))$ ($p \in [0, 1]$) w.r.t. a uniform distribution on the *curve parameter*:

$$L(\theta) = \mathbb{E}_{p \sim U(0,1)}[\ell(\phi_\theta(p))] \quad (1)$$

The generic curve function $\phi_\theta(p)$ has been characterized as a polygonal chain or Bezier curve in prior works [10, 4]. An example of the polygonal chain characterization can be seen in Equation 2. The bend in the curve is parameterized by θ , and the curve finding procedure minimizes Equation (1).

$$\phi_\theta(p) = \begin{cases} 2(p\theta + (0.5 - p)w_1), & 0 \leq p \leq 0.5 \\ 2((p - 0.5)w_2 + (1 - p)\theta), & 0.5 \leq p \leq 1 \end{cases} \quad (2)$$

The curve finding procedure begins by initializing θ as the midpoint of the line segment between w_1 and w_2 . Then, in each training round, we sample $\hat{p}$ from a uniform distribution, and a gradient step is applied to θ with respect to $\ell(\phi_\theta(\hat{p}))$. This effectively fixes the curve endpoints w_1 and w_2 while allowing the midpoint (i.e., parameters θ) to move in the weight space. To recover the

model parameters from each point of a trained curve, we first train a polygonal curve using the described procedure. Once we obtain the optimal midpoint parameters θ , we can recover the model parameters by setting a value for the curve parameter p and evaluating the curve at that point. For instance, if we set $p = 0.75$, then the point $\phi_\theta(0.75)$ corresponds to the midpoint of the line segment between θ and the original curve endpoint w_2 , which can be expressed as $0.5w_2 + 0.5\theta$. Consequently, the value at each dimension of the recovered model is equal to the average value of that dimension between w_2 and the optimal parameters θ^* .

We can use mode connectivity to study DNN model convergence, as it allows for a generic procedure in selecting the next minima without imposing strict conditions on its parameters. To prove convergence between local and global models in a DP framework, we outline three main transformations.

2 Convergence Analyses

We present a comprehensive convergence analysis for general DP-FL and UDP-FL. This analysis utilizes the notation listed in Appendix Table 1.

3 Mode Connectivity-based Convergence

We can utilize mode connectivity to study the convergence of deep neural network (DNN) models, as it provides a flexible procedure for selecting subsequent minima without imposing stringent conditions on the model parameters.

First, we introduce the worst-case estimate for the additional number of rounds required by UDP-FL. Then, we present an optimal strategy to accelerate convergence. Let w_t^k be the model parameter maintained in device k in training round t .

Theorem 1 (Proof in Appendix A). *Let w^* represent the model parameters at a local minimum. Suppose θ_t^k and $\tilde{\theta}_t^k$ are the parameters trained over the curve function $\phi_\theta(p)$ for the non-DP and DP models, respectively, where the DP model is perturbed by the Staircase mechanism with parameters Δ and ϵ . The system UDP-FL will transport $\tilde{\theta}_t^k$ to w^* with less than $O\left(\Delta^2 \cdot \frac{\epsilon^\epsilon}{(\epsilon^\epsilon - 1)^2}\right)$ additional rounds compared to the baseline federated learning (FL) for θ_t^k to reach w^* .*

Theorem 2. *The average of the optimal choice of parameter $\bar{\theta}_{t+1}$ which minimizes Equation 1 over a quadratic Bezier curve is*

$$\bar{\theta}_{t+1}^* = \frac{1.2}{L} + 1.1\bar{w}_t - 0.1\bar{v}_{t+1} \quad (3)$$

For a FedAvg algorithm trained on a quadratic Bezier curve with curve parameter r (equivalent to p in Section 1) randomly sampled from $U(0, 1)$, the

update of FedAvg with full clients active can be described as

$$w_{t+1}^k = (1-p)^2 v_{t+1}^k + 2(r-r^2)\theta_{t+1}^k + r^2 w_t^k, \quad (4)$$

where $v_{t+1}^k = w_t^k - \eta_t \nabla F_k(w_t^k, \xi_t^k)$ and θ_{t+1}^k is the Bezier function parameter which minimizes Equation 1.

Similar to [13, 9], we define two virtual sequences $\bar{v}_{t+1} = \sum_{k=1}^N p_k v_{t+1}^k$ and $\bar{w}_t = \sum_{k=1}^N p_k w_t^k$ where the former originates from an single step of SGD from the latter. Finally, it is clear that we always have

$$\bar{w}_{t+1} = (1-\bar{r})^2 \bar{v}_{t+1} + 2(\bar{r}-\bar{r}^2)\bar{\theta}_{t+1} + \bar{r}^2 \bar{w}_t$$

Proof. Suppose the step-size between the two sequential average model weights $\bar{w}_{t+1}$ and $\bar{w}_t$, i.e., $|w_{t+1}^k - w_t^k|$, is α . From the L-smoothness of F_k , it follows that

$$\begin{aligned} E_k \left\{ E_{\xi_{t+1}^k} (F_k(w_{t+1}^k, \xi_{t+1}^k)) \right\} &\leq E_k \left\{ E_{\xi_t^k} (F_k(w_t^k, \xi_t^k)) \right\} \\ &\quad - \left| \frac{\alpha^2 L}{2} - \alpha \right| \cdot E_k \left\{ \|\nabla F_k(w_t^k, \xi_t^k)\|_2^2 \right\} \end{aligned} \quad (5)$$

Let $\bar{F}(\bar{w}_t) = E_k \left\{ E_{\xi_t^k} (F_k(w_t^k, \xi_t^k)) \right\}$ and $\bar{\nabla} F(\bar{w}_t) = E_k \left\{ \nabla F_k(w_t^k, \xi_t^k) \right\}$, we can rewrite Equation (5) as follows.

$$\bar{F}(\bar{w}_{t+1}) \leq \bar{F}(\bar{w}_t) - \left| \frac{\alpha^2 L}{2} - \alpha \right| \cdot \bar{\nabla} F(\bar{w}_t) \quad (6)$$

A high-accuracy path between v_{t+1}^k and w_t^k leverages the parameters θ that minimize the expectation over a uniform distribution on the curve [6], i.e.,

$$\theta_{t+1}^{k*} = \arg \min_{\theta} \left\{ E_{r \sim U(0,1)} [F_k(w_{t+1}^k, \xi_{t+1}^k)] \right\} \quad (7)$$

From Equation (5), it follows that

$$\min_{\theta} \left\{ E_{r \sim U(0,1)} [\bar{F}(\bar{w}_{t+1})] \right\} \Leftrightarrow \max_{\theta} \left| \frac{\alpha^2 L}{2} - \alpha \right| \quad (8)$$

But the expression on the right hand side is strictly increasing w.r.t. α and its maxima occurs at $\alpha = \frac{1}{L}$. Therefore, we will solve the following problem.

$$\bar{\theta}_{t+1}^* = \arg \min_{\theta} \left| \alpha(\theta) - \frac{1}{L} \right| \quad (9)$$

From the expression for $\bar{w}_{t+1}$ it follows that

$$\alpha(\theta) - \frac{1}{L} = (1-\bar{r})^2 \bar{v}_{t+1} + 2(\bar{r}-\bar{r}^2)\bar{\theta}_{t+1} + (\bar{r}^2-1)\bar{w}_t - \frac{1}{L}$$

Therefore,

$$\begin{aligned} \min_{\theta} \left| \alpha(\theta) - \frac{1}{L} \right| &\Leftrightarrow \\ \min_{\theta} \left\{ \left| (1-\bar{r})^2 \bar{v}_{t+1} + 2(\bar{r}-\bar{r}^2)\bar{\theta}_{t+1} + (\bar{r}^2-1)\bar{w}_t - \frac{1}{L} \right| \right\} \end{aligned}$$

By rearranging the term inside the minimum function and substituting $\bar{r}^n$ by $\frac{1}{n}$ (refer to moments of uniform random variable defined over $[0, 1]$), we have $\bar{\theta}_{t+1}^*$ equal to

$$\begin{aligned} \arg \min_{\bar{\theta}_{t+1}} & \left| \bar{v}_{t+1} - \bar{w}_t - \frac{1}{L} + \bar{\theta}_{t+1} - \bar{v}_{t+1} + \frac{1}{12}(\bar{v}_{t+1} + \bar{w}_t - 2\bar{\theta}_{t+1}) \right| \\ & = \arg \min_{\bar{\theta}_{t+1}} \left| -\frac{1}{L} + \frac{5}{6}\bar{\theta}_{t+1} + \frac{1}{12}\bar{v}_{t+1} - \frac{11}{12}\bar{w}_t \right| \end{aligned}$$

Finally, by forcing the term inside the absolute value to be zero we must have

$$\bar{\theta}_{t+1}^* = \frac{1.2}{L} + 1.1\bar{w}_t - 0.1\bar{v}_{t+1}.$$

Thus, this completes the proof. $\square$

As a result, the average loss is minimized if all clients set their mode connectivity parameter to $\theta_{t+1}^k = \frac{1.2}{L} + 1.1w_t^k - 0.1v_{t+1}^k$. Alternatively, if the updates are subjected to clipping, FedAvg can maintain a fixed step size of $\frac{1}{L}$. These constraints are integrated into our UDP-FL; see Algorithm 1 in Appendix A.

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A Proof of Theorem 1

Proof. We will prove that if the original system converges in K rounds the expectation of the number of rounds $\tilde{K}$ needed for $\tilde{\theta}_k$ to approach 1 ($w_n \rightarrow w$) increases by $O(\sigma^2)$ where for the staircase noise $\sigma = \Delta \cdot \frac{e^{\epsilon/2}}{e^{\epsilon}-1}$ [7]. We'll assume that the non-perturbed version converges in K rounds and find k in terms of K . Since UDP-FL on client machines exerts the mode connectivity algorithm when searching for optima, we need to investigate diffusion mechanism of the perturbed parameter $\tilde{\theta}$ and investigating how it affects the curve function $\phi_{\tilde{\theta}}(p)$.

Given the perturbed gradient update:

$$\tilde{\theta}_{t+1} = \tilde{\theta}_t - \eta(\nabla_{\theta} L(\tilde{\theta}_t) + \xi_t)$$

where ξ_t is the noise added by the staircase mechanism. The curve function with perturbed parameter $\tilde{\theta}$ is:

$$\phi_{\tilde{\theta}}(p) = \begin{cases} 2(p\tilde{\theta} + (0.5 - p)w_1), & 0 \leq p \leq 0.5 \\ 2((p - 0.5)w_2 + (1 - p)\tilde{\theta}), & 0.5 \leq p \leq 1 \end{cases}$$

The loss function evaluated at the curve with perturbed parameter:

$$L(\tilde{\theta}) = \mathbb{E}_{p \sim U(0,1)}[\ell(\phi_{\tilde{\theta}}(p))]$$

Let's analyze the expected number of rounds k for the noisy parameter $\tilde{\theta}_k$ to approach the target value (similar to the convergence of θ in K rounds). Gradient Descent with Noise:

$$\tilde{\theta}_{t+1} = \tilde{\theta}_t - \eta(\nabla_{\theta} L(\tilde{\theta}_t) + \xi_t)$$

We know the non-perturbed algorithm converges in K rounds. For the noisy version, the convergence will be slower due to the added noise. The noise impact can be understood by examining how the noise affects the variance of the parameter updates over iterations. Let's denote the variance of the noise ξ_t as σ^2 . The variance of the parameter updates accumulates over iterations:

$$\text{Var}(\tilde{\theta}_k) = \sum_{t=1}^k \eta^2 \sigma^2$$

The noise introduces a diffusion term that slows down convergence, i.e., the number of iterations k needs to compensate for this diffusion. In the following we will show that extra term is bounded by $\frac{C\sigma^2}{\eta^2}$ where C is a constant that depends on the specific noise characteristics and the gradient landscape. The proof requires the following assumptions on the clients' loss functions $\{F_k\}$ (standard assumption [8]).

Assumption 1. $\{F_k\}'s$ are L -smooth: $F_k(v) \leq F_k(w) + (v - w)^T \nabla F_k(w) + \frac{L}{2} \|v - w\|_2^2, \forall v, w$.

Assumption 2. $\{F_k\}'s$ are μ -strongly convex: $F_k(v) \geq F_k(w) + (v - w)^T \nabla F_k(w) + \frac{\mu}{2} \|v - w\|_2^2, \forall v, w$.

Assumption 3. Let ξ_t^k be sampled from the k -th device's data uniformly at random. The variance of stochastic gradients in each device is bounded: $E \|\nabla F_k(w_t^k, \xi_t^k) - \nabla F_k(w_t^k)\|^2 \leq \sigma_k^2, \forall k$.

Assumption 4. The expected squared norm of stochastic gradients is uniformly bounded, i.e., $E \|\nabla F_k(w_t^k, \xi_t^k)\|^2 \leq G^2, \forall k, t$.

Based on the findings from [8], the FedAvg algorithm's updates with only a subset of devices active can be explained as follows: Let w_t^k represent the model parameters on the k -th device at time step t , and let $\mathcal{I}_E$ denote the set of global synchronization steps, defined as $\mathcal{I}_E = \{nE \mid n = 1, 2, \dots\}$. If $t + 1$ is in $\mathcal{I}_E$, indicating a communication round, FedAvg engages all devices. For partial device participation, the updates proceed as:

$$\begin{aligned} v_k &= w_k - \eta \nabla F(w_k, \xi_k), \quad \text{if } t + 1 \notin \mathcal{I}_E, \\ w_{t+1} &= \sum_{k=1}^N p_k v_k, \quad \text{if } t + 1 \in \mathcal{I}_E. \end{aligned}$$

Here, the variable v_k represents the immediate result of a single SGD step from w_k . The parameter w_{t+1}^k is viewed as the result after synchronization steps. In this context, we define two aggregated sequences: $v_t = \sum_{k=1}^N p_k v_t^k$ and $w_t = \sum_{k=1}^N p_k w_t^k$. The sequence v_{t+1} is derived from performing one SGD step on w_t . When $t + 1 \notin \mathcal{I}_E$, both sequences are unavailable. If $t + 1 \in \mathcal{I}_E$, we can access w_{t+1} . For simplicity, we define $g_t = \sum_{k=1}^N p_k \nabla F_k(w_t^k)$ and $\bar{g}_t = \sum_{k=1}^N p_k \nabla F_k(w_t^k, \xi_t^k)$. Consequently, $v_{t+1} = w_t - \eta g_t$ and $E[g_t] = g_t$. Assuming that Assumptions 1 and 2 hold, and if $\eta_t \leq 1$, the following inequality is satisfied:

$$\begin{aligned} E[\|v_{t+1} - w^*\|^2] &\leq (1 - \eta_t \mu) E[\|w_t - w^*\|^2] \\ &\quad + \eta_t^2 E[\|\bar{g}_t - g_t\|^2] + 6L\eta_t^2 \Gamma + 2E[\|w_t - w_t^k\|], \end{aligned}$$

where $\Gamma = F^* - \sum_{k=1}^N p_k F_k^* \geq 0$.

Assuming convergence of non-perturbed model when v_{t+1} follows the mode connectivity path (along θ_t), the path of UDP-FL will follow

$$\begin{aligned} E[\|\tilde{\theta}_{t+1} - w^*\|^2] &\leq (1 - \eta_t \mu) E[\|\tilde{\theta}_t - w^*\|^2] \\ &\quad + \eta_t^2 E[\|g_t - \bar{g}_t\|^2] + 6L\eta_t^2 \Gamma + 2E[\|\tilde{\theta}_t - \tilde{\theta}_t^k\|], \end{aligned}$$

where $\Gamma = F^* - \sum_{k=1}^N p_k E_{p \sim U(0,1)}[\ell(\phi_{\tilde{\theta}_t^k}(p))] \geq 0$. Only first term and last term will generate additional term when moving from non-perturbed to the perturbed path. Additional terms in both terms are variance of the noisy residue ζ_t which is $\Delta^2 \cdot \frac{e^\epsilon}{(e^\epsilon - 1)^2}$. $\square$

Algorithm 1: Mode Connectivity Curve Finding Procedure

Input: w_1, w_2 : Model parameters for two local minima

Output: Optimal curve parameters θ

- 1 Initialize θ as the midpoint: $\theta = 0.5(w_1 + w_2)$
- 2 Define loss function $\ell(w)$ (e.g., cross-entropy loss)
- 3 Define curve function $\phi_\theta(p)$:

$$\phi_\theta(p) = \begin{cases} 2(p\theta + (0.5 - p)w_1) & \text{for } 0 \leq p \leq 0.5, \\ 2((p - 0.5)w_2 + (1 - p)\theta) & \text{for } 0.5 < p \leq 1. \end{cases}$$

- 4 **while** θ *not converges* **do**
 - 5 Sample $\hat{p}$ uniformly from $[0, 1]$
 - 6 Compute loss: $L(\theta) = \ell(\phi_\theta(\hat{p}))$
 - 7 Compute gradient: $\nabla_\theta L(\theta)$
 - 8 Update θ using a gradient step: $\theta \leftarrow \theta - \eta \nabla_\theta L(\theta)$
 - 9 **return** *Optimal parameters* θ
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B On Convergence Assumptions

The convexity of cross-entropy is elucidated by examining its formulation in terms of Kullback-Leibler (KL) divergence: $H(p, q) = H(p) + D_{KL}(p||q)$. As $H(p)$ remains constant, attention is directed to the convex nature of the KL divergence component. Notably, KL divergence exhibits convexity for discrete pairs (p, q) , as demonstrated by (p_1, q_1) and (p_2, q_2) [12]. The proof of this convexity, utilizing the log-sum inequality[14], is provided here [11], offering a comprehensive understanding of KL divergence’s convexity within the context of cross-entropy. Additionally, Berrada et al. [2] established the smoothness of cross entropy.

Table 1: Notations for convergence analysis.

Symbol	Description
N	the number of clients
T	number of global epochs
p_k	the weight of the k -th device
n_k	the number of training data samples in the k -th device
w_t^k	the latest model on the k -th device
η_t	the decaying learning rate at round t
v_t^k	the immediate result of one step SGDUpdate from w_t^k
w^*	the model which minimizes F
ξ_t^k	a sample uniformly chosen from the local data
$F(\cdot)$	the weighted average loss
$F_k(\cdot)$	a user-specified loss function
$F_k^*(\cdot)$	the minimum of $F_k(\cdot)$
F^*	the minimum of F
σ_k^2	the upper bound variance of stochastic gradients in the k -th device
G	the upper bound expected squared norm of stochastic gradients in k -th device
μ	the strongly convex constant of $F_k(\cdot)$
L	the smoothness constant of $F_k(\cdot)$
θ	mode connectivity parameter
r	mode connectivity percentage